

A disproof of Conjecture 00000008197

Symmetry breaking of Weierstrass gap semigroups already happens in genus 2

1 The conjecture

Conjecture 00000008197 reads:

Definition: The gap distribution of computed Weierstrass points: the distribution $wgap(g)$ of Weierstrass gap sequences of curves. *Conjecture:* The multiplicity of the gaps (the number of symmetric gaps) is at most $g - 1$ (a symmetry upper bound); the minimal genus of symmetry breaking of gaps (nonsymmetric semigroups) is $g = 3$ (a third-genus breaking); the realization density of semigroups (the semigroup distribution of random curves, realizing all numerical semigroups in genus g) concentrates on symmetric semigroups (a symmetry concentration law), with concentration $1 - 2^{-g}$ (a concentration formula); and the realization frequency of nonsymmetric semigroups at large g is $2^{-g}g^c$ up to polynomial correction (a nonsymmetric polynomial correction).

This is a conjunction of five clauses: (1) the symmetry upper bound, (2) third-genus breaking, (3) the symmetry concentration law, (4) the concentration formula $1 - 2^{-g}$, and (5) the nonsymmetric polynomial correction. A Weierstrass gap sequence at a point P of a smooth projective curve of genus g is the complement of the semigroup $H(P)$ of pole orders at P , which is a numerical semigroup of genus g . The conjecture itself speaks of “nonsymmetric semigroups” and of “realizing all numerical semigroups in genus g ”, so its objects are numerical semigroups.

- **Main reading.** $wgap(g)$ ranges over the numerical semigroups of genus g (all of them realized, as the conjecture states). Clause (2) is then false: the least genus of a nonsymmetric numerical semigroup is 2, witnessed by $\langle 3, 4, 5 \rangle$ (Theorem 6).
- **Weierstrass-point reading.** Only semigroups at Weierstrass points proper are recorded, i.e. the non-ordinary ones. Clause (4) is then false at $g = 2$: every such semigroup of genus 2 is the symmetric $\langle 2, 5 \rangle$, so every distribution gives the symmetric semigroups mass $1 \neq 1 - 2^{-2}$ (Theorem 8).

In both cases the conjunction is false, whatever the other clauses mean.

2 Definitions

We follow Rosales and Garcia-Sanchez, *Numerical Semigroups* (Springer, 2009), Ch. 1 and 3, and the Wikipedia page “Numerical semigroup” (https://en.wikipedia.org/wiki/Numerical_semigroup), which says: “The number of elements in the set of gaps $G(S)$ is called the genus of S ” and “A numerical semigroup S is symmetric if and only if $g(S) = (F(S) + 1)/2$.”

Definition 1. A *numerical semigroup* is an additive submonoid $S \subseteq \mathbb{N}$ with finite complement (Lean: `NumericalSemigroup`, fields `carrier` : `AddSubmonoid` $\mathbb{N}$ and `finite_gaps`). Its *gaps* are $G(S) = \mathbb{N} \setminus S$ (`gaps`). Its *genus* is $g(S) = |G(S)|$ (`genus`, via `Set.ncard`). Its *Frobenius number* $F(S)$ is the largest gap (`IsFrobenius` S F := `IsGreatest` S .`gaps` F).

Definition 2. S is *symmetric* (`IsSymmetric`) if either $S = \mathbb{N}$ (so $F = -1$, and $g = (F + 1)/2 = 0$), or $F(S) - x \in S$ for every gap x (Rosales and Garcia-Sanchez, Cor. 4.5: S is symmetric iff $x \in \mathbb{Z} \setminus S$ implies $F(S) - x \in S$; for $x < 0$ this is automatic).

Definition 3. The *ordinary* semigroup of genus g has gaps $\{1, \dots, g\}$ (`IsOrdinary`). The Wikipedia page “Weierstrass point” (https://en.wikipedia.org/wiki/Weierstrass_point) states: “The gap sequence is $1, 2, \dots, g$ for a non-Weierstrass point. For a Weierstrass point it contains at least one higher number.” It also states: “Multiplication of functions gives the non-gaps a numerical semigroup structure”. So the semigroups that occur at Weierstrass points of genus- g curves are non-ordinary numerical semigroups of genus g (`WeierstrassType` g).

3 Clause (2): third-genus breaking fails

Lemma 4 (`S345_gaps`, `S345_genus`, `S345_not_symmetric`). $S_{345} = \langle 3, 4, 5 \rangle = \{0\} \cup \{3, 4, 5, \dots\}$ is a numerical semigroup with gaps $\{1, 2\}$, genus 2 and Frobenius number 2. It is not symmetric: the gap $x = 1$ has $F - x = 1 \notin S$. Equivalently (`S345_genus_ne`), $2g = 4 \neq 3 = F + 1$.

Proof. A sum of two elements of $\{0\} \cup [3, \infty)$ is 0 or at least 3. The rest is direct. $\square$

S_{345} is the ordinary semigroup of genus 2 (`S345_isOrdinary`). It is the semigroup at every non-Weierstrass point of every curve of genus 2, so it is realized.

Lemma 5 (`one_mem_gaps`, `isSymmetric_of_genus_lt_two`). If $g(S) \geq 1$ then 1 is a gap. Every numerical semigroup of genus 0 or 1 is symmetric.

Proof. If $1 \in S$ then every $n = n \cdot 1 \in S$ and $g = 0$. Genus 0 means $S = \mathbb{N}$. Genus 1 means $G(S) = \{1\}$, so $F = 1$, and $F - 1 = 0 \in S$. $\square$

Theorem 6 (`least_nonsymmetric_genus`, `not_thirdGenusBreaking`). The least genus of a nonsymmetric numerical semigroup is 2: *IsLeast nonsymmetricGenera* 2. Hence *ThirdGenusBreaking* (*IsLeast nonsymmetricGenera* 3) is false.

4 Clause (4) at genus 2 under the Weierstrass-point reading

Lemma 7 (`gaps_of_genus_two`). Every numerical semigroup of genus 2 has gap set $\{1, 2\}$ or $\{1, 3\}$.

Proof. 1 is a gap. If 2 and 3 were both in S , then every $n \geq 2$ would be in S (induction $n \mapsto n + 2$, `mem_of_two_three`), so $G(S) \subseteq \{1\}$ and $g \leq 1$. Hence $\{1, 2\}$ or $\{1, 3\}$ is contained in $G(S)$, and both sides have two elements. $\square$

Theorem 8 (`isSymmetric_of_weierstrassType_two`, `symmetric_mass_eq_one`, `no_distribution_with_concentration_formula`). Every non-ordinary numerical semigroup of genus 2 has gaps $\{1, 3\}$, i.e. it is $\langle 2, 5 \rangle$, and it is symmetric ($F = 3$, with $3 - 1 = 2 \in S$ and $3 - 3 = 0 \in S$). Consequently, for every probability mass function μ on *WeierstrassType* 2,

the symmetric semigroups have mass 1. Hence there is no distribution with symmetric mass $1 - 2^{-2} = 3/4$: *ConcentrationFormula 2* is false. The type is nonempty (*S25_weierstrass*), so this is not vacuous.

Remark 9 (Checks and other readings). A Python enumeration of all numerical semigroups of genus ≤ 8 reproduces the counts 1, 1, 2, 4, 7, 12, 23, 39, 67 (OEIS A007323). It confirms that the least nonsymmetric genus is 2. It also shows that the symmetric share under the uniform distribution $(1, 1/2, 1/2, 3/7, \dots)$ for $g = 1, 2, 3, 4$ is never $1 - 2^{-g}$ for $1 \leq g \leq 8$. Clause (1) uses the undefined notion “number of symmetric gaps” and plays no role here.

5 Lean formalization

File `lean/Conjecture8197/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1; only `import Mathlib`). The main theorem is

```
theorem conjecture_00000008197_false (P Q : Prop) :
  ¬(ThirdGenusBreaking ∧ P) ∧ ¬(ConcentrationFormula 2 ∧ Q) ∧
  (S345.genus = 2 ∧ ¬S345.IsSymmetric) ∧ IsLeast nonsymmetricGenera 2 ∧
  Nonempty (WeierstrassType 2) ∧
  (∀μ : PMF (WeierstrassType 2), μ.toOuterMeasure{S | S.1.IsSymmetric} = 1)
```

where *ConcentrationFormula g* is $\exists \mu : \text{PMF } (\text{WeierstrassType } g), \mu.\text{toOuterMeasure}\{S \mid S.1.\text{IsSymmetric}\} = 1 - (2^{-1})^g$. The propositions *P* and *Q* stand for any formalization of the remaining clauses.

Axiom audit. `#print axioms Conjecture8197.conjecture_00000008197_false` reports `[propxt, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`, and the file also compiles with `-DwarningAsError=true`.